

ALI ALAA SALAH

Machine Learning Engineer | Data Analyst

alialaasalah43@gmail.com | linkedin.com/in/ali-alaa-salah | Egypt

PROFESSIONAL SUMMARY

Computer Science student focused on Machine Learning and Data Analysis, with practical experience through academic, personal, and training projects. Skilled in Python, SQL, Pandas, NumPy, Scikit-learn, and PyTorch, with hands-on work in data preprocessing, exploratory data analysis, machine learning, computer vision, deep learning, and recommendation systems. Currently developing industry-ready skills through the Digital Egypt Pioneers Initiative (DEPI) Machine Learning Engineer track.

EDUCATION

Alexandria Private University — Faculty of Computer Science

Bachelor's Degree in Computer Science | Expected Graduation: 2028

TRAINING & PROJECTS

Digital Egypt Pioneers Initiative (DEPI) — Machine Learning Engineer Track | 2026

Training in Python, Mathematics for Machine Learning, Data Science, Machine Learning, Deep Learning, NLP, Computer Vision, Prompt Engineering, and Azure AI Fundamentals.

Hybrid Movie Recommendation System | Python, Pandas, NumPy, Scikit-learn, Surprise

- Built a hybrid recommendation approach combining collaborative filtering and content-based filtering.
- Implemented SVD collaborative filtering and cosine-similarity-based content recommendations.
- Evaluated the SVD model using RMSE and MAE; achieved RMSE 0.8727 and MAE 0.6703.

COVID-19 Radiography Image Classification | Python, PyTorch, Computer Vision

- Performed EDA and preprocessing on a multi-class chest X-ray dataset covering COVID, Normal, Viral Pneumonia, and Lung Opacity.
- Applied image normalization, augmentation, stratified data splitting, and learning-curve analysis.
- Compared deep learning architectures using validation performance and training behavior.

Stroke Prediction & Healthcare Data Analysis | Python, Pandas, NumPy, Scikit-learn

- Performed data cleaning, exploratory analysis, preprocessing, feature preparation, and classification modeling.
- Evaluated model performance using appropriate classification metrics.

Face Recognition System | Python, Computer Vision, Deep Learning

- Developed a face recognition project involving image preprocessing, resizing, tensor conversion, and model evaluation.

Hotel Management Database | SQL, Database Design

- Designed a relational database for guests, rooms, bookings, and hotel operations using SQL and relational database concepts.

Pharmacy Management System | Java, OOP

- Developed a Java application for pharmacy management and applied object-oriented programming principles.

Smart Car Racing — AI Driver | Python, Artificial Intelligence

- Developed an AI-based racing project focused on environment interpretation and autonomous driving decisions.

Sudoku Solver | Programming, Algorithms

- Implemented an algorithmic Sudoku solver using logical constraints and search-based problem solving.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Data Cleaning, EDA, Data Visualization

Machine Learning: Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Cross-Validation, Overfitting & Underfitting

Deep Learning & Computer Vision: PyTorch, Neural Networks, Image Classification, Image Preprocessing, Data Augmentation, Face Recognition

Recommendation Systems: Collaborative Filtering, Content-Based Filtering, Hybrid Recommendation Systems, SVD, Cosine Similarity

Tools: Jupyter Notebook, Google Colab, VS Code, Git/GitHub fundamentals

SOFT SKILLS

Problem Solving • Analytical Thinking • Teamwork • Communication • Attention to Detail • Continuous Learning • Project Documentation

CERTIFICATIONS & COURSES

Digital Egypt Pioneers Initiative (DEPI) — Machine Learning Engineer Track | 2026

Additional coursework: Python, Data Science, Machine Learning, Deep Learning, NLP, Computer Vision, Prompt Engineering, Mathematics for Machine Learning, Azure AI Fundamentals.

LANGUAGES

Arabic: Native | **English:** Professional Working Proficiency

CAREER INTERESTS

Machine Learning Engineering • Data Science • Data Analysis • Computer Vision • Artificial Intelligence • Recommendation Systems